

Business Statement & Executive Summary

Business Statement :

The objective of this project is to analyze e-commerce sales and customer transaction data using SQL to identify key revenue and profitability drivers, understand customer purchasing behavior, evaluate product and regional performance, and uncover areas requiring business attention. The analysis provides actionable insights into sales trends, customer retention, product profitability, and state-category performance to support data-driven business decisions.

Business Problems Solved

1. Overall revenue and profit
2. Monthly revenue and profit
3. Month-over-month revenue growth
4. Product category sales volume
5. Category contribution to revenue
6. Low-profit / loss-making subcategories
7. Top 10 customers by spending
8. Repeat customer rate
9. RFM customer segmentation
10. Top 5 states by order volume
11. Average Order Value by state
12. Most frequently used payment mode
13. Product assortment
14. Running revenue total
15. High-sales but low-margin State × Category combinations

Executive Summary :

This project analyzes e-commerce sales and transaction data using SQL to evaluate revenue, profitability, customer behavior, product performance, and regional sales patterns. The analysis measures overall and monthly revenue/profit, category contribution, product profitability, customer spending, repeat-purchase behavior, RFM customer rankings, payment preferences, and state-level performance. Advanced SQL techniques including JOINS, CTEs, subqueries, NTILE(), LAG(), and window functions were used to identify trends and performance patterns. The analysis also identifies state-category combinations with above-

average sales but below-average profit margins, helping highlight areas where the business may need to review pricing, costs, or product strategy.